

REST API Specification & Endpoint Manual

Complete Technical OpenAPI Reference & Request/Response Specification Catalog

Document Title:	REST API Specification Manual
API Base Protocol:	FastAPI REST JSON (OpenAPI 3.0 Specification)
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Security Scheme:	OAuth2 Bearer JWT Token (HS256 Encryption)

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1.0 DOCUMENT REVISION HISTORY

Rev #	Date	Author	Reviewer	Approver	Description of Changes
1.0	2026-08-01	Narendar R.	Manvitha	Infosys Mentor	Initial Baseline API Endpoints Reference
1.5	2026-08-04	Ashwindh	Pravallika	Technical Lead	Added openpyxl Export & OTP Reset Endpoints
2.0	2026-08-07	Team PricePilot	QA Team	Academic Committee	Final OpenAPI 3.0 Production Validation

2.0 GLOBAL API ARCHITECTURE & STATUS CODES

HTTP Code	Status Name	Functional Scenario / Description
200 OK	Success	Request processed successfully; returns JSON object or file binary.
201 Created	Resource Created	User registration or new database resource successfully created.
400 Bad Request	Client Error	Invalid input parameters, duplicate email/username, or malformed JSON.
401 Unauthorized	Auth Failure	Missing or expired OAuth2 Bearer JWT access token.
403 Forbidden	Permission Denied	Authenticated user lacks Admin privileges required for route.
404 Not Found	Resource Missing	Target document, user ID, or endpoint URL does not exist.
422 Validation Error	Unprocessable Entity	Pydantic schema validation failure on request payload.
500 Internal Error	Server Error	Unhandled exception in FastAPI service layer.
503 Service Error	Service Unavailable	ML prediction model (.pkl) is uninitialized or loading.

3.0 COMPLETE REST API ENDPOINT CATALOG

Method	Endpoint Path Path	Auth Required	Router Module	Functional Purpose
POST	/api/auth/register	Public	auth.py	Register new user account with Bcrypt password hash.
POST	/api/auth/login	Public	auth.py	Authenticate credentials and issue JWT access token.
POST	/api/auth/forgot-password	Public	auth.py	Generate and store 6-digit password reset OTP.
POST	/api/auth/verify-otp	Public	auth.py	Verify 6-digit OTP code and update user password.
GET	/api/auth/me	Bearer JWT	auth.py	Return active authenticated user profile details.
POST	/api/predict	Bearer JWT	predict.py	Execute Extra Trees ML price prediction engine.
GET	/api/model-status	Public	predict.py	Check Extra Trees model loaded status & feature count.
GET	/api/dashboard/stats	Bearer JWT	dashboard.py	Fetch system analytics dashboard summary stats.
GET	/api/dashboard/recent-activity	Bearer JWT	dashboard.py	Fetch recent system audit activity logs.
GET	/api/users	Admin JWT	users.py	List all registered users (supports status filter).
POST	/api/users	Admin JWT	users.py	Admin creation of new user account.
PUT	/api/users/{user_id}	Admin JWT	users.py	Admin update of user status, role, and details.
DELETE	/api/users/{user_id}	Admin JWT	users.py	Admin deletion of user account.
POST	/api/users/bulk-status	Admin JWT	users.py	Bulk update status for multiple user accounts.
POST	/api/users/bulk-delete	Admin JWT	users.py	Bulk delete multiple user accounts.
GET	/api/admin/export-users	Admin JWT	users.py	Download openpyxl styled Users_Report.xlsx binary.
GET	/api/docs	Bearer JWT	docs.py	List available project PDF & PPTX document binaries.
GET	/api/docs/{doc_id}	Bearer JWT	docs.py	Get metadata preview for specific document.
GET	/api/docs/download/{doc_id}	Bearer JWT	docs.py	Download actual PDF or PPTX binary file.
GET	/api/health	Public	main.py	System health check & database status.

4.0 DETAILED ENDPOINT SPECIFICATIONS

4.1 POST /api/predict — Execute AI Price Prediction

- **Purpose:** Computes dynamic price recommendations, profit margins, and confidence scores using Extra Trees ML model.
- **Authentication:** OAuth2 Bearer JWT Token (`Authorization: Bearer `).
- **Validation Schema:** Pydantic `ProductFeatures` model enforcing positive numeric inputs.

Sample cURL Request:

```
curl -X POST 'http://localhost:8000/api/predict' \ -H 'Authorization: Bearer ' \ -H 'Content-Type: application/json' \ -d '{"order_item_id": 1, "freight_value": 25.50, "product_category_name": 5, "product_weight_g": 1200, "product_volume": 3500}'
```

Sample Success Response (200 OK):

```
{ "prediction_id": 142, "predicted_price": 189.50, "confidence_score": 0.965, "prediction_time": 0.045, "model_name": "Extra Trees Regressor", "demand_level": "High Demand", "profit_margin": 35.0, "estimated_cost": 123.18 }
```

4.2 GET /api/admin/export-users — openpyxl Excel Export

- **Purpose:** Streams native `.xlsx` workbook of user registry with dark blue headers (#1E3A8A), bold white text, borders, auto-adjusted column widths, and freeze panes.
- **Authentication:** Admin OAuth2 Bearer JWT Token (`role == 'Admin'`).
- **Response Header:** `Content-Disposition: attachment; filename=Users\_Report.xlsx`.

Sample cURL Request:

```
curl -X GET 'http://localhost:8000/api/admin/export-users' \ -H 'Authorization: Bearer ' \ --output Users_Report.xlsx
```

4.3 POST /api/auth/forgot-password & verify-otp — OTP Password Reset

- **Purpose:** Generates 6-digit OTP stored in `password\_reset\_otps` table with 15-minute expiration for secure password recovery.

Sample Request JSON (`POST /api/auth/verify-otp`):

```
{ "email_or_phone": "user@example.com", "otp_code": "482910", "new_password": "SecurePass123!" }
```

5.0 SECURITY NOTES & OPENAPI SCHEMA

All API endpoints enforce strict CORS origin validation, parameter sanitization via Pydantic v2, and SecurityHeadersMiddleware injection.

OpenAPI 3.0 JSON specification is published at `/openapi.json` and interactive Swagger UI at `/docs`.